

Training Configurations for Synthetic pre trained weights(Models trained on webots images)

Multi Class

Dataset

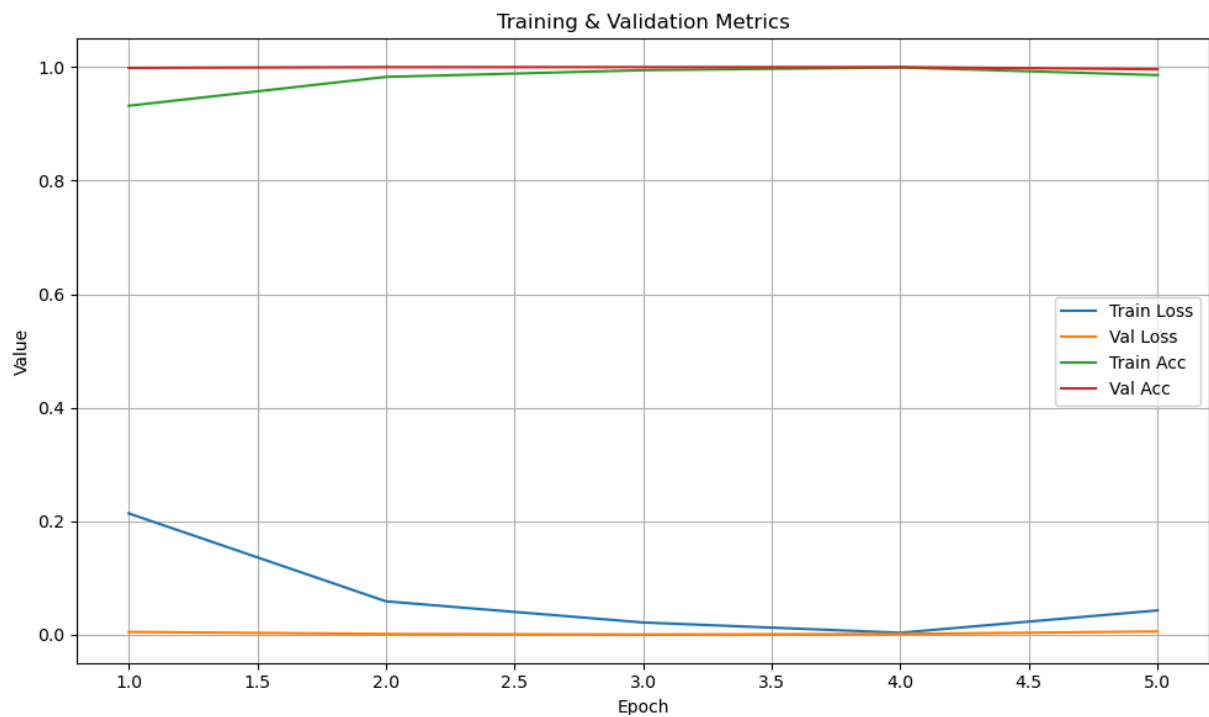
- Dataset type: Multi-class image classification
- Number of classes: 7
- Input image type: Grayscale (converted to 3-channel for ResNet)
- Image size: 224×224
- Data loading: Shuffled each epoch

Transforms / Preprocessing

- Resize(224×224)
- ToTensor()
- Normalize(mean = 0.5, std = 0.5)
- Grayscale images duplicated across 3 channels to fit ResNet50 input

Training Hyperparameters

- Epochs: 30
- Batch size: 16
- Optimizer: Adam
 - Learning rate: 1×10^{-4}
- Loss function: CrossEntropyLoss (multi-class)
- Model Architecture - ResNet50



Multi Label

Dataset & Labels

- Dataset type: Multi-label image classification
- Label encoding: Folder name encodes binary vector
 - Example: 1_0_1 $\rightarrow$ label = [1, 0, 1](left_right_forward)
- Number of labels: 3
- Image format: Grayscale $\rightarrow$ expanded to 3-channel
- Input size: 224×224
- Shuffling: Enabled per epoch (shuffle=True)
- num_workers: 2

Transforms (Preprocessing)

- Resize to 224×224
- Convert to tensor
- Normalize grayscale values with mean = 0.5, std = 0.5
- Repeat grayscale tensor across 3 channels

Training Hyperparameters

- Loss function: BCEWithLogitsLoss
- Optimizer: Adam
 - Learning rate: 1×10^{-4}
- Batch size: 16
- Epochs: 30
- Device: GPU if available (cuda)

Evaluation Metric

- Exact Match Accuracy
 - Converts logits $\rightarrow$ probabilities using sigmoid
 - Threshold: 0.5
 - A prediction is correct only if *all* labels match exactly

Training Configurations for Multi Label Training(Models trained on real-world data)

Dataset

- **Task type:** Multi-label image classification
- **Label encoding:** Directory names represent binary vectors
 - Example: "1_0_1" $\rightarrow$ [1, 0, 1]

- **Number of labels:** 3
- **Image type:** Grayscale, replicated to 3 channels
- **Input resolution:** 224×224
- **Dataset split:** 5-fold K-Fold Cross-Validation (shuffle=True, random_state=42)

Data Augmentation & Preprocessing

- Resize(224×224)
- ColorJitter(brightness=0.2, contrast=0.2)
- ToTensor()
- Normalize(mean=0.5, std=0.5)
- Grayscale repeated to RGB (img.repeat(3,1,1))

Training Setup (per fold)

- **Optimizer:** Adam
 - Learning rate: 1×10^{-3}
- **Loss:** BCEWithLogitsLoss (multi-label)
- **Batch size:** 16
- **Epochs:** up to 30
- **Device:** GPU (cuda) when available
- **Early Stopping:**
 - Patience: 3 epochs
 - Minimum improvement (delta): $1e-4$

Evaluation Metrics

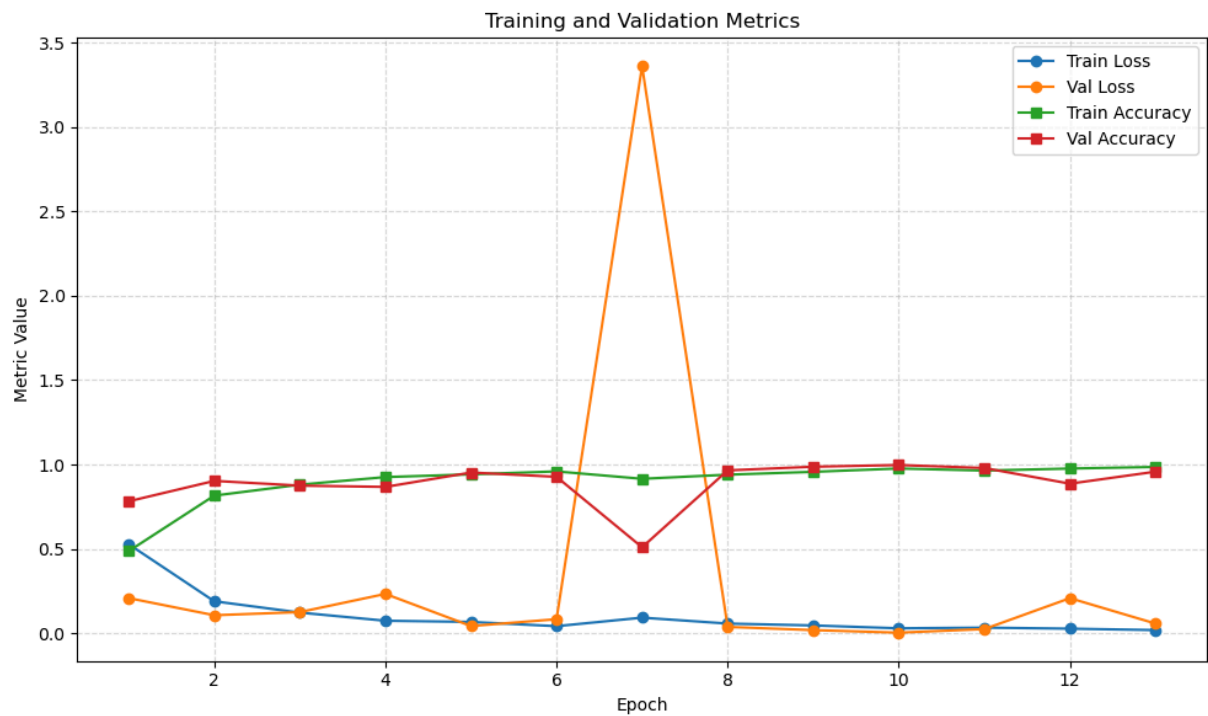
- **Primary metric:** *Exact multi-label accuracy*
 - Thresholding: $\text{sigmoid} > 0.5$
 - Uses sklearn's `accuracy_score` on binary vectors
- Metrics tracked per epoch:
 - Training loss
 - Validation loss
 - Training accuracy
 - Validation accuracy

Cross-Validation

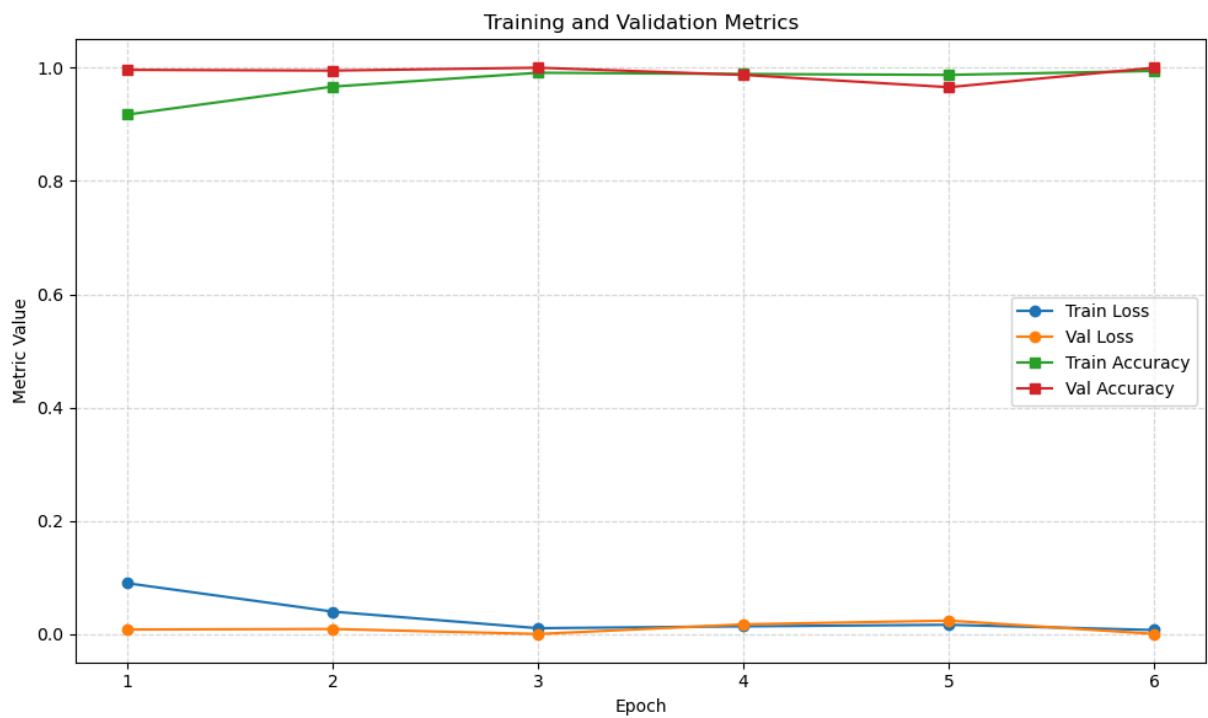
- Number of folds: 5
- Best validation accuracy recorded for each fold
- Final model = weights of fold with highest validation accuracy
- Reported results:
 - Mean accuracy across all folds
 - Best fold accuracy

Training and Validation Curves(Only the highest validation performance model) for ResNet50

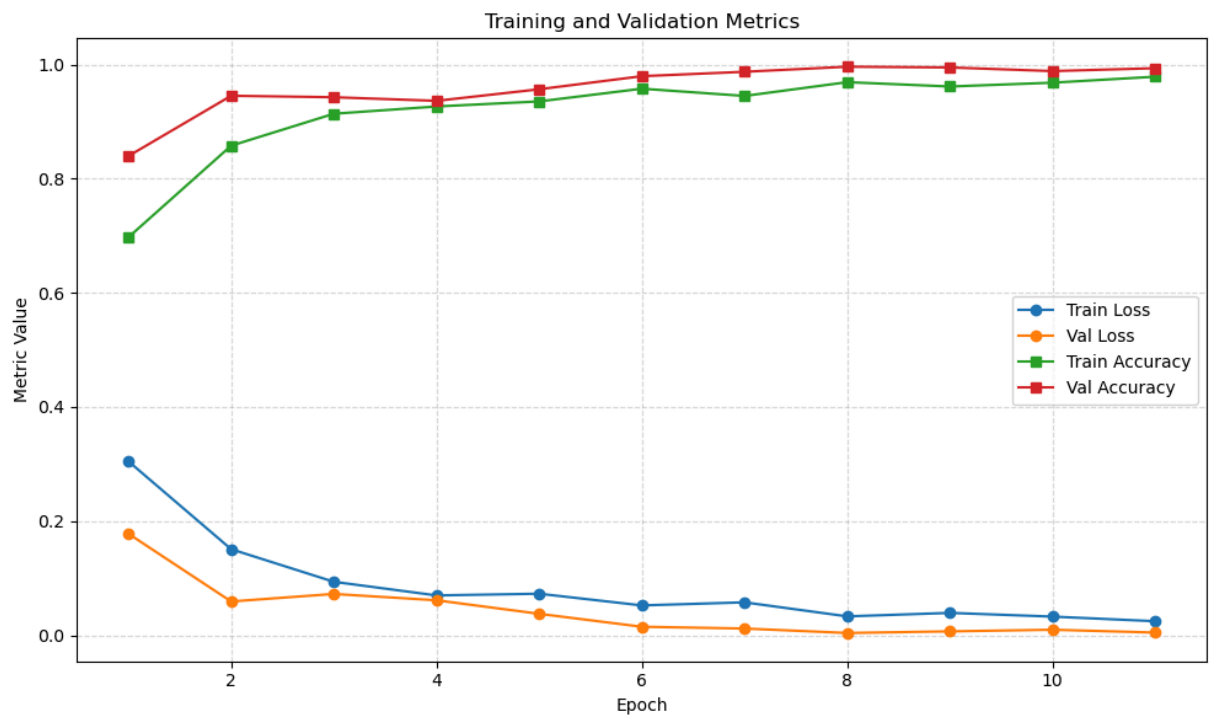
He Initialization



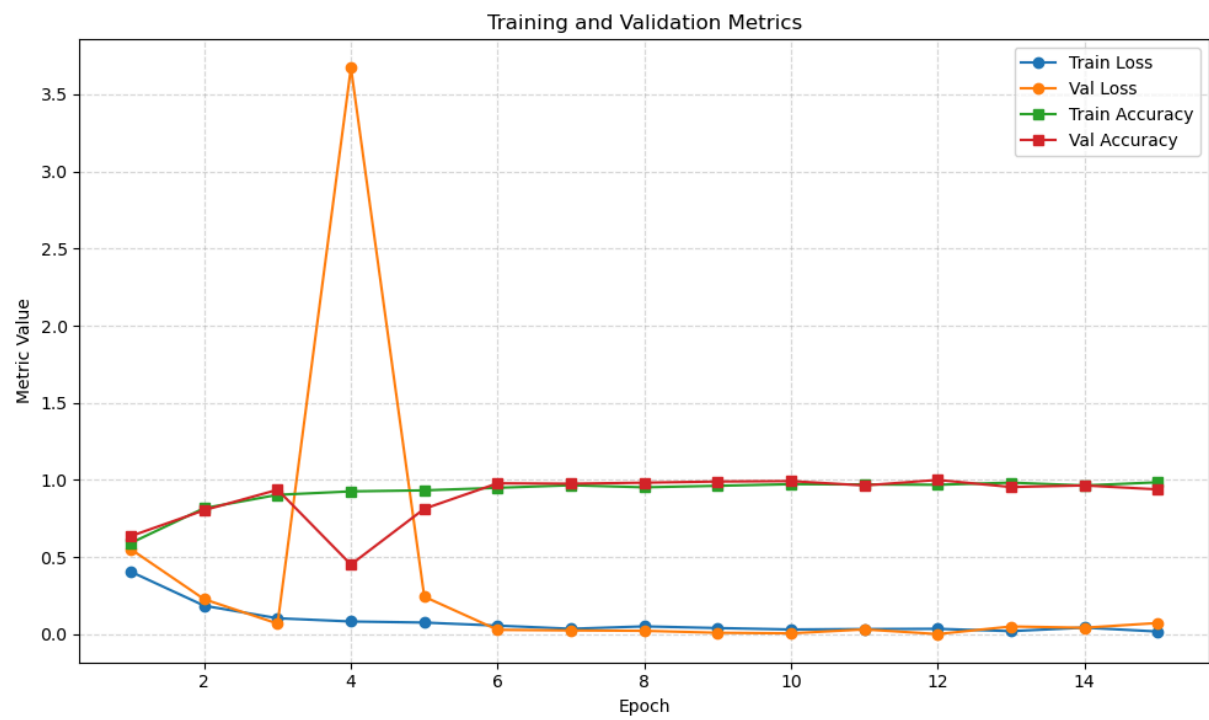
ImageNet Initialization



Synthetic(last 2) Initialization

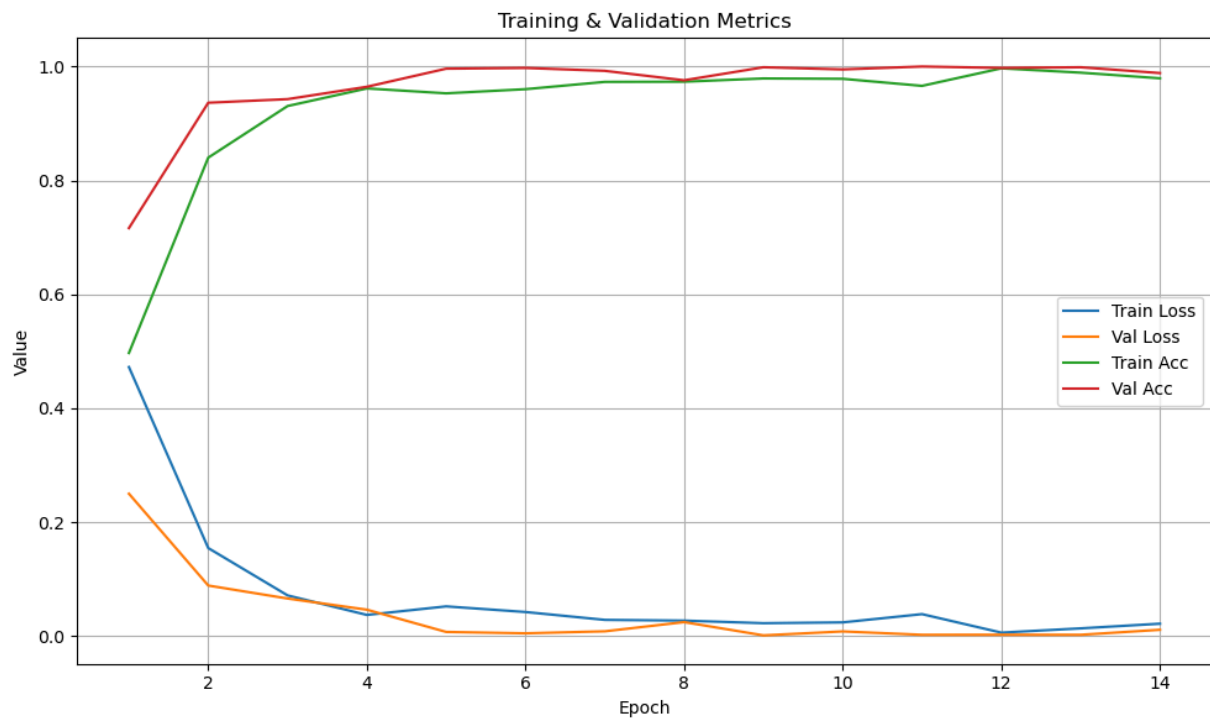


Synthetic(all) Initialization

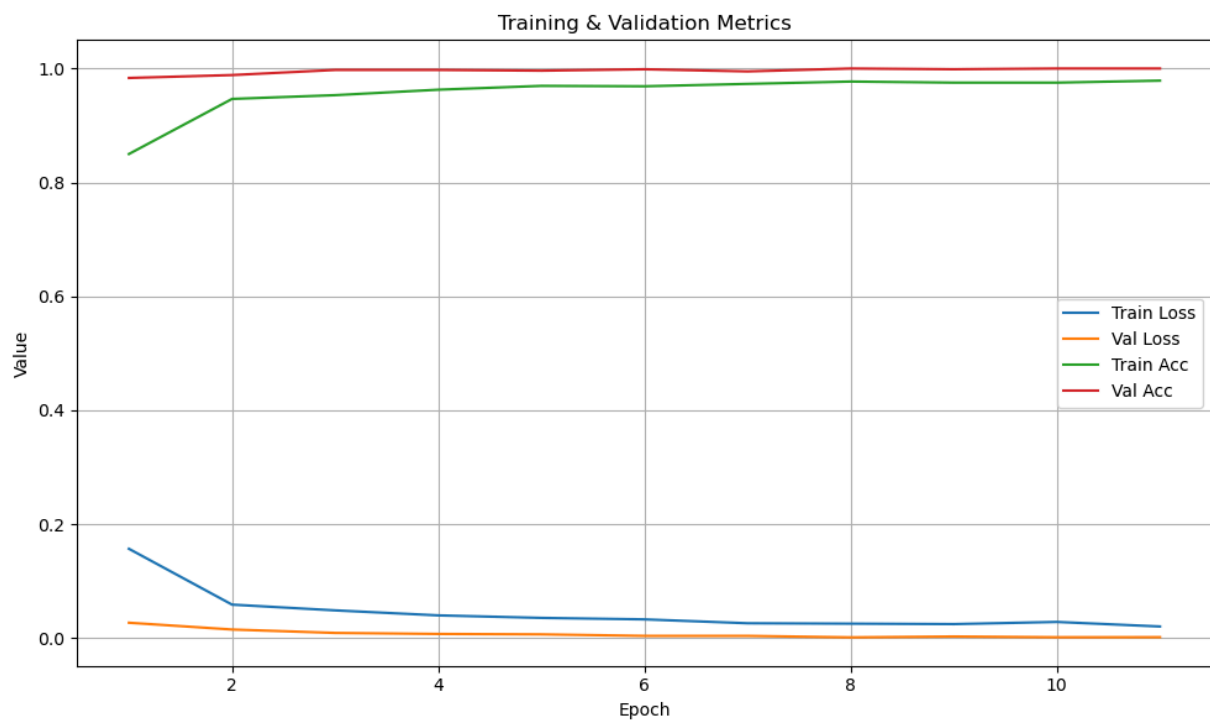


Training and Validation Curves(Only the highest validation performance model) for MobileNetV2

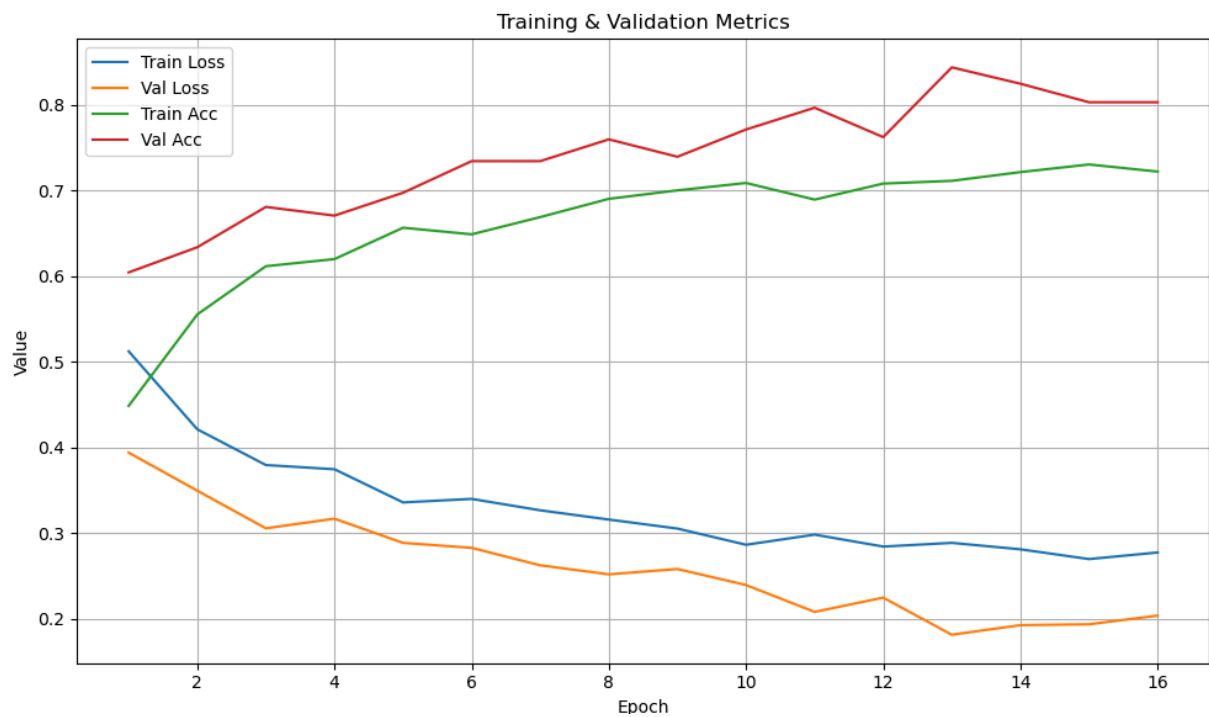
He Initialization



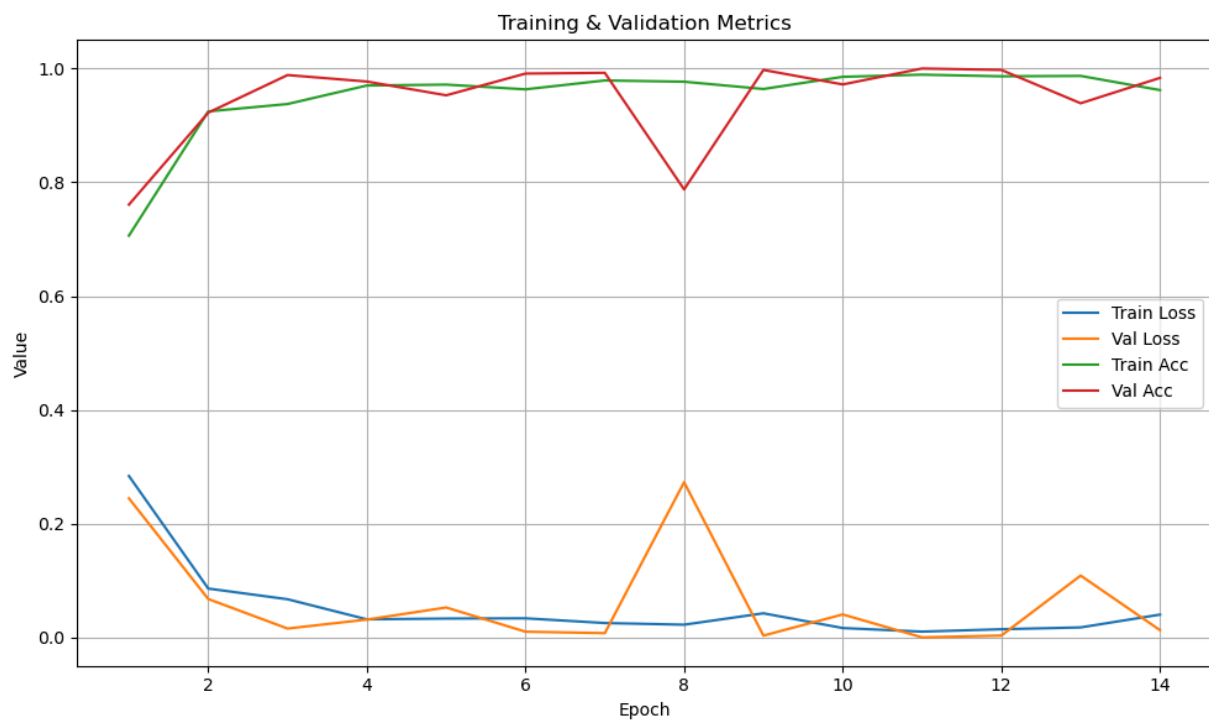
ImageNet Initialization



Synthetic(last 2) Initialization



Synthetic(all) Initialization



Grad-CAM Configurations for Explainability(Multi Label)

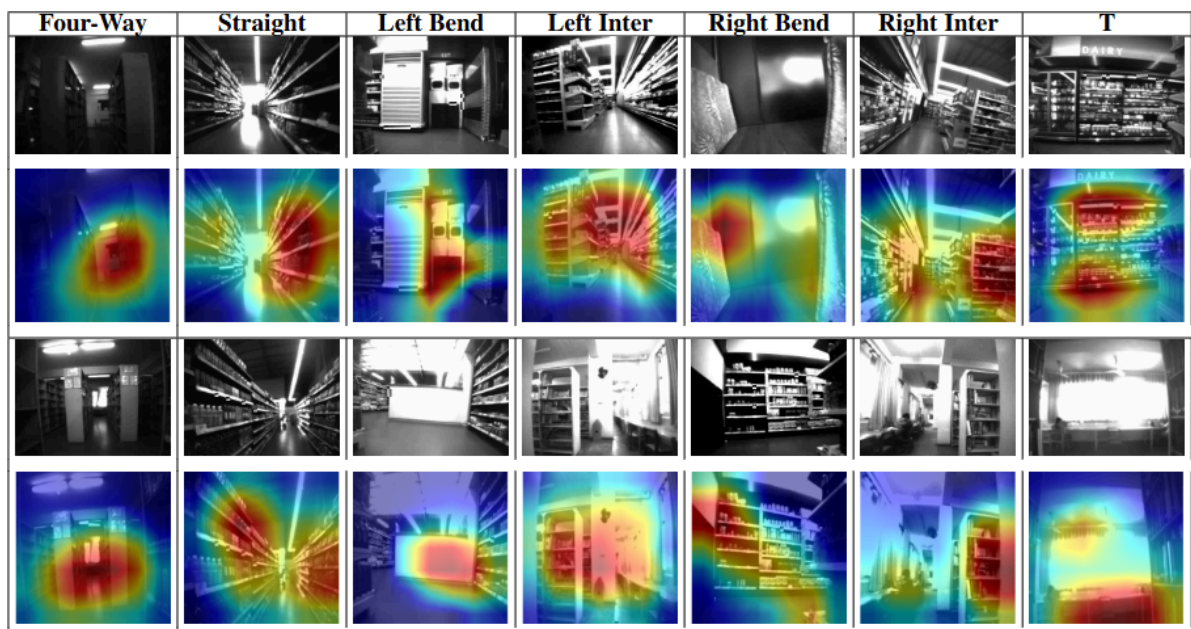
Target Layers (Grad-CAM)

- ResNet-50:
 - layer4[-1] (final Bottleneck block)
- MobileNetV2:
 - features[-1] (final convolutional block)

Input Preprocessing

- Images are loaded in grayscale (convert('L'))
- Resized to 224×224
- Normalization:
 - Mean: 0.5
 - Std: 0.5
- Grayscale expanded to 3-channel (by repeating the channel)
- Final tensor shape for each image: [1, 3, 224, 224]

Multi class configuration: ResNet50



Multi Label Configuration

TABLE 5.3: Grad-CAM visualization results for multi-label configuration on the He and ImageNet Settings on three different samples(Resnet50)

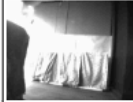
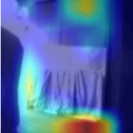
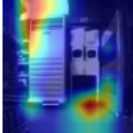
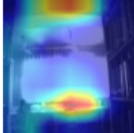
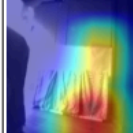
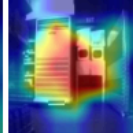
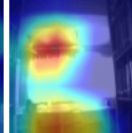
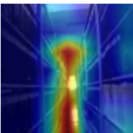
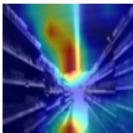
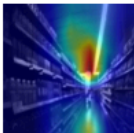
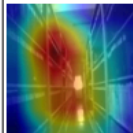
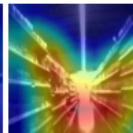
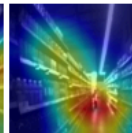
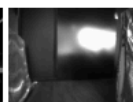
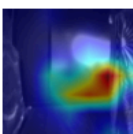
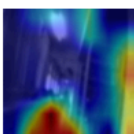
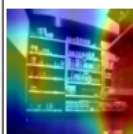
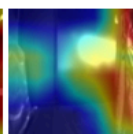
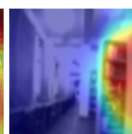
Prominent Label	Kaiming He			ImageNet		
Left Original						
Left Grad-CAM						
Forward Original						
Forward Grad-CAM						
Right Original						
Right Grad-CAM						

TABLE 5.4: Grad-CAM visualization results for multi-label configuration on the Digital Twin Setting for three different samples(Resnet50)

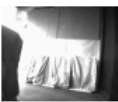
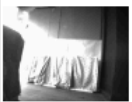
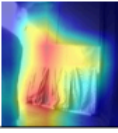
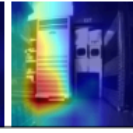
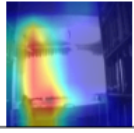
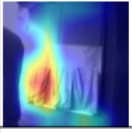
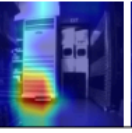
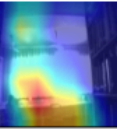
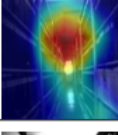
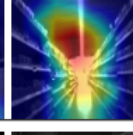
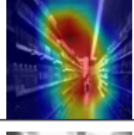
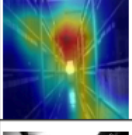
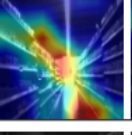
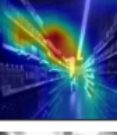
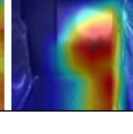
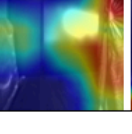
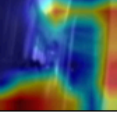
Prominent Label	Digital Twin(Last 2)			Digital Twin(All)		
Left Original						
Left Grad-CAM						
Forward Original						
Forward Grad-CAM						
Right Original						
Right Grad-CAM						

TABLE 5.6: Grad-CAM visualization results for multi-label configuration on the He and ImageNet settings using three different samples (MobileNetV2).

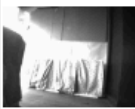
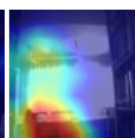
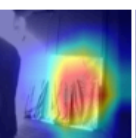
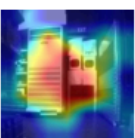
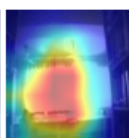
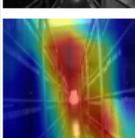
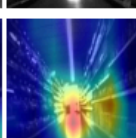
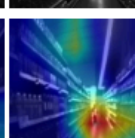
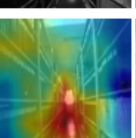
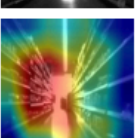
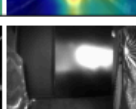
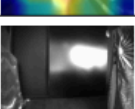
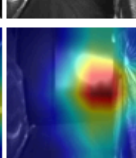
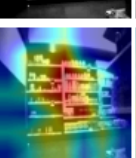
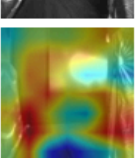
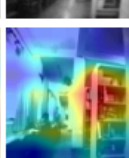
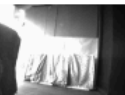
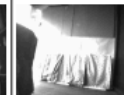
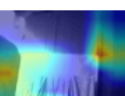
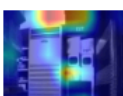
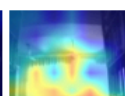
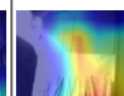
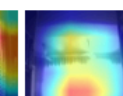
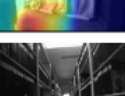
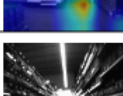
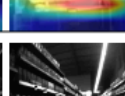
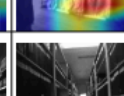
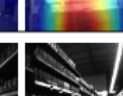
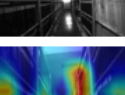
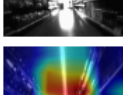
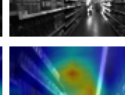
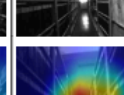
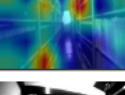
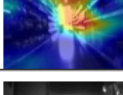
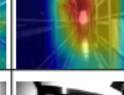
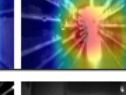
Prominent Label	Kaiming He			ImageNet		
Left Original						
Left Grad-CAM						
Forward Original						
Forward Grad-CAM						
Right Original						
Right Grad-CAM						

TABLE 5.7: Grad-CAM visualization results for multi-label configuration on the Digital Twin setting for three different samples (MobileNetV2).

Prominent Label	Digital Twin (Last 2)			Digital Twin (All)		
Left Original						
Left Grad-CAM						
Forward Original						
Forward Grad-CAM						
Right Original						
Right Grad-CAM	